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Particle-flow topology optimization with the SPH kernel for hinge-free designs of compliant mechanisms using isogeometric material point method

Jiayi Zhu[a,d], Daji Lin[b], Liang Gao[b], Huai Cao[c], Jie Gao[a,d,*]

[a] Department of Engineering Mechanics, School of Aerospace Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China
[b] State Key Lab of Digital Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan, 430074, China
[c] School of Design, Huazhong University of Science and Technology, Wuhan, 430074, China
[d] Hubei Key Laboratory for Engineering Structural Analysis and Safety Assessment, Huazhong University of Science and Technology, Wuhan 430074, China

Abstract

Compliant mechanisms, which achieve input-output functions through continuous elastic deformation, offer inherent advantages such as frictionless motion, absence of backlash, and suitability for microscale fabrication. These features render them highly promising for advanced engineering applications. However, traditional topology optimization approaches, typically relying on fixed solid material distributions, often produce localized geometric hinges at theoretically optimal configurations, thereby limiting manufacturability and compromising operational reliability. To overcome these challenges, this paper presents a novel particle-flow topology optimization (PFTO) framework grounded in a Lagrangian-Eulerian hybrid formulation. By explicitly incorporating particle mobility and interactions, the proposed method enhances the adaptability of compliant mechanisms to complex loading and deformation scenarios encountered in practical engineering. The approach introduces two key innovations: (1) the application of a smoothed particle hydrodynamics (SPH) kernel function for weighted smoothing of particle properties, effectively suppressing numerical instabilities arising from irregular particle distributions and insufficient local interactions; (2) the facilitation of free particle migration and self-organization during iterative optimization, which eliminates conventional fixed-geometry hinges and enables the structure to naturally accommodate large deformations and dynamic loading conditions. Numerical examples involving both two- and three-dimensional compliant mechanisms demonstrate the superior performance and robustness of the PFTO method, underscoring the critical role of particle motion in avoiding undesirable hinge formation and improving structural functionality.

Keywords: Topology optimization, Compliant mechanisms, Particle-flow, Isogeometric material point method

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1 Introduction

A compliant mechanism is a structure that relies on the elastic deformation of the material itself to achieve motion and force transmission [1]. Compared with traditional rigid mechanisms, it has significant advantages such as no friction, no gap, and a compact structure [2]. Due to these characteristics, Compliant mechanisms have been widely used in high-precision fields such as aerospace precision instruments, medical devices, and micro-electromechanical systems [3,4]. With the continuous improvement of requirements for structural performance, reliability, and lightweight, traditional empirical design methods have been unable to meet the design requirements under complex working conditions [5]. Especially when faced with the highly coupled and nonlinear characteristics of compliant mechanisms, it is urgent to develop a systematic, numerically driven topology optimization method to achieve its optimal performance design.

In the design of Compliant mechanisms, topology optimization, as an important means of structural optimization, provides a systematic design path for achieving high-performance, lightweight and high-reliability mechanisms [6,7] The core idea of topology optimization is to automatically find the optimal material distribution within the design domain through mathematical modeling and numerical iteration, thereby revealing innovative structural configurations [8,9]. Current topology optimization methods can be roughly divided into two categories: grid-based topology optimization methods [10] and geometry-based topology optimization methods [11,12]. The basic idea of grid-based methods, such as the SIMP method, is to optimize the structural morphology by allocating material density on a fixed background grid and gradually adjusting the material distribution [13]. Geometry-based topology optimization methods, such as the Level Set Method (LSM) [14],

*Corresponding author.
E-mail address: JieGao@hust.edu.cn

Moving Morphable Components (MMC) [15], and the derivative method moving deformable cavities (MMV) [16], leverage parameterized geometric descriptions to update the boundaries of the design domain flexibly. These methods directly manipulate the evolution of geometric boundaries, effectively avoiding the generation of artifact elements while maintaining boundary clarity and geometric continuity. They also demonstrate significant advantages in achieving consistency in multiscale structural design and manufacturing. Consequently, geometry-driven topology optimization has emerged as a key development direction in flexible mechanism optimization research in recent years, providing new theoretical and technical support for achieving higher-precision, higher-performance mechanism design [17,18].

However, due to its discrete characteristics, extremely soft areas with single-node connections are often generated at the corners of the mesh units, forming the so-called "factual hinge" problem, which affects the continuity and manufacturability of the mechanism [19]. To address this issue, researchers have proposed a variety of solutions: Based on the grid density method, Zhu et al. [20] proposed a weighted multi-objective function of average input-output flexibility and a SIMP adaptive weight iteration strategy to alleviate the hinge phenomenon effectively. However, the method depends on parameter settings and is affected by the grid scale, which limits the geometric expression ability. For coupled-field devices, Gao et al. [21] proposed robust isogeometric topology optimization (RITO), which further combines the SIMP method with three-field density mapping (erosion-intermediate-dilation) and NURBS geometry. It can explicitly eliminate single-node hinges in the optimization and maintain uniform manufacturability. It is suitable for multi-physics scenarios such as piezoelectric actuators, but it requires separate evaluation of multiple "states", which increases the computational cost. In the geometry-based level set method (LSM), Guo et al. [22] implemented explicit feature size constraints on signed distance functions to achieve direct control of the minimum/maximum length scale, thereby mechanistically reducing the generation of hinge and thin neck features; The explicit geometry MMC/MMV framework uses deformable components as design primitives [23], naturally carrying geometric parameters such as thickness. It can control scale and connectivity while maintaining smooth boundaries to reduce the risk of hinges [24]. However, it is complex to implement, has a high computational cost, and has limited adaptability to initial conditions and multi-physics field coupling. The "bubble method" improves coherence and manufacturability by adaptively introducing/removing cavities and facilitating shape evolution, and suppresses degraded connections in compliant mechanisms. However, its topological diversity is limited under high degrees of freedom and complex loads [25]. To overcome the limitations of the above methods in terms of continuity and geometric representation, Samaniego et al [26] proposed a deep energy method with thermodynamic and variational consistency. This method does not require traditional mesh discretization and can directly solve direct problems and optimization problems. It naturally considers system uncertainties in the modeling process and can combine experimental data to achieve physically consistent optimization and inverse design. Although this method is essentially a meshless framework, it still uses NURBS for geometric representation to maintain the smoothness and high-order continuity of the model. In addition, this method has also been applied in the context of parameter mapping in Refs. [27] and [28], and has high implementation simplicity and versatility.

Hughes proposed Isogeometric Analysis (IGA) and his team [29,30], and can be regarded as an extension of the classical finite element method in numerical analysis. In the past few decades, many studies have explored the application of IGA in topology optimization [31,32]. Early on, research focused on improving the discrepancy between geometric modeling and structural optimization in traditional finite element analysis using the IGA method [33]. Subsequently, Xu et al. [34] proposed an isogeometric topology optimization design model for flexible mechanisms and verified the effectiveness of this method in solving topological discontinuities and structural boundary problems. Li et al. [35] proposed a planar compliant mechanism design method based on an isogeometric topology optimization framework. By integrating geometric representation and sensitivity calculation, the optimal deformation path of structures such as flexible hinges was accurately identified, and the probability of single-node connections was reduced. At present, to improve numerical accuracy and eliminate numerical artifacts in the material point method (MPM) [36], IGA has also been introduced into the MPM [37,38]. Moutssanidis et al. [38] proposed a new method based on MPM and IGA called IGA-MPM, which achieves high-order smooth approximation by utilizing NURBS. and Lin et al. [39] proposed a PFTO framework based on I-MPM combined with optimization. The numerical discussion clearly reveals that the PFTO method has key features such as particle motion, two mappings of physical information, and two inverse mappings of sensitivity information [40]. Based on this, the characteristics of PFTO provide a solution to eliminate hinge problems for topological optimization of flexible mechanisms. However, the current research on the application of PFTO in flexible mechanisms is very limited, and its theoretical system and numerical implementation are still in the exploratory stage.

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This paper proposes a particle-flow topology optimization method described by Lagrangian-Eulerian under the isogeometric framework suitable for Compliant mechanisms. This method introduces a movable particle flow in the design domain, transforms the design variables of the traditional fixed control points into the evolution problem of particle positions and properties, thereby realizing continuous and controllable adjustment of material distribution. The particle motion is described by Lagrangian-Eulerian hybrid [41], which enhances the responsiveness to structural topological changes while maintaining geometric continuity [42,43]. Compared with traditional methods, this strategy not only retains the high-precision advantage of isogeometric analysis, but also demonstrates unique design potential in the optimization of hingeless flexible mechanisms. The main features of the proposed particle-flow topology optimization method are as follows:

(1) The mobility of particles enables the material to be flexibly redistributed in the design domain, significantly improving the adaptability of the Compliant mechanism to displacement targets and deformation paths. Compared with the traditional optimization method with fixed control points, the free movement of particles enhances the expression ability and design freedom of topological evolution.

(2) The smoothness of MPM effectively suppresses numerical oscillation and maintains the continuity of physical quantities through information mapping between particles and background grids. This feature is particularly critical for large deformation analysis in Compliant mechanisms, which helps to improve the stability of the optimization process and the reliability of the results.

In summary, this paper focuses on the topological optimization design of compliant mechanisms based on particle flow and isogeometric methods. First, this paper first proposes that there is a hinge problem in the Compliant mechanism, that is, the current optimization model based on statics drive tends to guide the rigid link structure, thereby naturally generating point flexibility to realize the motion function. For details, see Section 2. In Section 3, the equal geometry material point method (I-MPM) based on smooth high-order discretization is used to solve the elastic force balance equation, which effectively overcomes the numerical instability problem existing in the application of the traditional finite element method in the material point method, and constructs a particle flow model based on Lagrangian-Euler description. Section 4 further proposes the corresponding topology optimization formula to maximize the single-point output displacement. In Section 5, the numerical implementation process is given, and the effectiveness and applicability of the proposed PFTO method are verified through multiple examples.

2 The design of compliant mechanism

In the past few decades, since the landmark paper by Sigmund [44], continuous topology optimization methods for compliant mechanisms (CMs) have been proposed in various forms [45]. This diversity is mainly due to the differences in design requirements—people usually want the designed mechanism to have different functions, such as transmitting force, outputting displacement, or transmitting energy [46,47]. Fig. 1(a) shows the compliant mechanism synthesis problem, whose optimization goal is to maximize the amplitude of the target output displacement $\boldsymbol{f}_{\text{in}}$ under the action of the input load $\boldsymbol{d}_{\text{out}}$. Since the magnitude of the input load does not affect the optimization results of the structure considered in this study, $\boldsymbol{f}_{\text{in}}$ is set to unit force.

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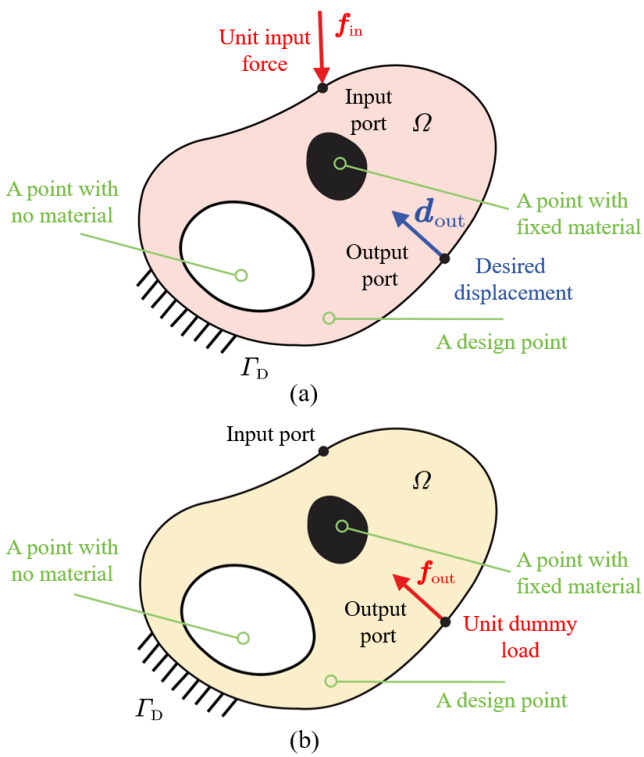

Fig. 1 Typical problems of compliant mechanism synthesis: (a) Load case 1, (b) Load case 2.

where $\Omega \subseteq \boldsymbol{R}^2$ is an open bounded set occupied by linear elastic structures. $\boldsymbol{\Gamma_D}$ represents the Dirichlet boundary. The input force $\boldsymbol{t}$ is applied to the input port $\boldsymbol{I}$ and will generate an output $\boldsymbol{O}$ at the output port. Usually, a spring with a stiffness of $\boldsymbol{k}_0$ is connected to the output port to simulate the reaction force from the workpiece. This is the well-known spring model [6,48]. To simplify the problem, most literature uses a linear spring model and expresses the output port force as:

$$\boldsymbol{F_o} = \boldsymbol{k_o}\Delta_o. \tag{1}$$

Among them, $\boldsymbol{k_o}$ is the equivalent stiffness of the output end, and Δ_o is the displacement. Experiments have shown that when the constraints of this model are not introduced, the optimized structure often cannot achieve an effective connection between the input and output ports, making it difficult to form a functional mechanism. Therefore, although this model is only a simplification, it has become the mainstream method in the topology optimization of compliant mechanisms [49].

However, it is difficult to accurately evaluate the coupling effect between the input and output from an energy perspective using only the unidirectional load input model shown in Fig. 1(a). To this end, a virtual load model is introduced as shown in Fig. 1(b). In this model, a unit dummy load $\boldsymbol{f}_{\text{out}}$ is applied to the output end, and its direction is consistent with the target output displacement. In this way, the original goal of "maximising the output displacement" can be equivalently transformed into maximising the mutual compliance $\boldsymbol{C}_{\text{mut}}$ between the input and output loads, which is defined as:

$$\boldsymbol{C}_{\text{mut}} = \int_{\boldsymbol{W}} \boldsymbol{s}\left(\boldsymbol{u_{out}}\right) : \boldsymbol{e}\left(\boldsymbol{u_{in}}\right) \mathrm{d}\boldsymbol{W}, \tag{2}$$

where, $\boldsymbol{u}_{\text{in}}$ and $\boldsymbol{u}_{\text{out}}$ denotes the displacement field caused by the input force $\boldsymbol{f}_{\text{in}}$ and the output dummy load $\boldsymbol{f}_{\text{out}}$, respectively. Then, the two components of input and output can be expressed as:

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$$\begin{aligned} \boldsymbol{C}_{\text{in}} &= \frac{1}{2}\int_{W} \boldsymbol{s}\left(\boldsymbol{u}_{in}\right) : \boldsymbol{e}\left(\boldsymbol{u}_{in}\right) \mathrm{d}\boldsymbol{W}, \\ \boldsymbol{C}_{\text{out}} &= \frac{1}{2}\int_{W} \boldsymbol{s}\left(\boldsymbol{u}_{out}\right) : \boldsymbol{e}\left(\boldsymbol{u}_{out}\right) \mathrm{d}\boldsymbol{W}. \end{aligned} \tag{3}$$

This problem can be modeled as a topology optimization problem of a linear elastic structure under equilibrium conditions. Its weak form is as follows

$$\int_{\boldsymbol{\Omega}} \boldsymbol{E}_{ijL_q} \boldsymbol{\varepsilon}_{ij}\left(\boldsymbol{u}\right) \boldsymbol{\varepsilon}_{L_q}\left(\boldsymbol{v}\right) \mathrm{d}\boldsymbol{\Omega} = \int_{\boldsymbol{\Omega}} \boldsymbol{f} \times \boldsymbol{v} \mathrm{d}\boldsymbol{\Gamma}, \quad \boldsymbol{u}|_{\Gamma_D} = 0, \forall \boldsymbol{v} \in \boldsymbol{U}, \tag{4}$$

where $\boldsymbol{\varepsilon}$ is the strain tensor, $\boldsymbol{f}$ is the external force, $\boldsymbol{u}$ is the displacement field, and $\boldsymbol{U}$ is the allowed test function space. In addition, the design area is restricted by the total volume constraint:

$$\int_{\boldsymbol{\Omega}} \mathrm{d}\boldsymbol{\Omega} \le \boldsymbol{V}_{\max} \tag{5}$$

and impose a maximum limit on the input port displacement:

$$\Delta_{\boldsymbol{i}} \le \Delta_{\max}. \tag{6}$$

3 Isogeometric Lagrangian-Euler description

3.1 Isogeometric Material Point Method (I-MPM)

The Material Point Method (MPM) is a numerical method based on Lagrangian description that can be used to simulate the static response of continuous media. Its basic concept is to discretize an object into a series of material points, each of which carries state variables such as stress and volume (or mass). In the MPM, a reference configuration is selected, and the initial state of the object is described in material coordinates. The current configuration after deformation is then transformed from the reference configuration to spatial coordinates using a mapping function.

In static analysis, the system no longer includes inertial terms and explicit time derivatives. The change in variables with t simply represents the progression of the load step or iteration step, describing the gradual application of external loads or boundary conditions—that is, the continuous evolution of the structure from an unloaded state to an equilibrium state. By introducing a field φ defined throughout the entire object, the changes in physical quantities at material points under different load steps can be tracked, thereby providing a unified description of the spatial distribution and update mechanism of various field variables within the static MPM framework. This function-field transformation has two definitions, corresponding to different description methods:

- Lagrangian description: Directly using the material coordinates $\boldsymbol{X}$ and the load step $\boldsymbol{t}$ as independent variables, the change of the field quantity is only determined by the gradual evolution of the properties of the material point itself with the load, and its partial derivative is defined as

$$\frac{\boldsymbol{D}\varphi_t\left(\boldsymbol{X}\right)}{\boldsymbol{Dt}} \equiv \dot{\varphi} = \partial_t \varphi_t\left(\boldsymbol{X}\right). \tag{7}$$

From this perspective, the focus is on the movement and evolution of the particles themselves, such as how the position, density, and strain of a material point gradually adjust in response to external forces. This description embodies the essential idea of the material point method—that each particle carries complete physical information and moves with deformation.

- Euler description: This field is defined at the spatial position $\varphi_t\left(\boldsymbol{x}^n\left(\boldsymbol{X}\right)\right)$, which not only considers the time change but also includes the influence of the spatial position changing with the deformation map, so the partial derivative is

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$$\frac{D\varphi_t(\boldsymbol{x})}{Dt} = \partial_t \varphi_t(\boldsymbol{x}) + \nabla \varphi_t(\boldsymbol{x}) \times \boldsymbol{V}_t(\boldsymbol{X}), \qquad (8)$$

where $\boldsymbol{V}_t$ represents the velocity field, which represents the velocity of the material point in the reference domain at the current moment. The Euler perspective emphasizes the changes in physical quantities at fixed spatial points. Changes in the density values or physical quantity distributions of grid control points are reflected as an Euler update: the control points themselves do not move, but their corresponding density, stress, or displacement fields are gradually adjusted with particle mapping.

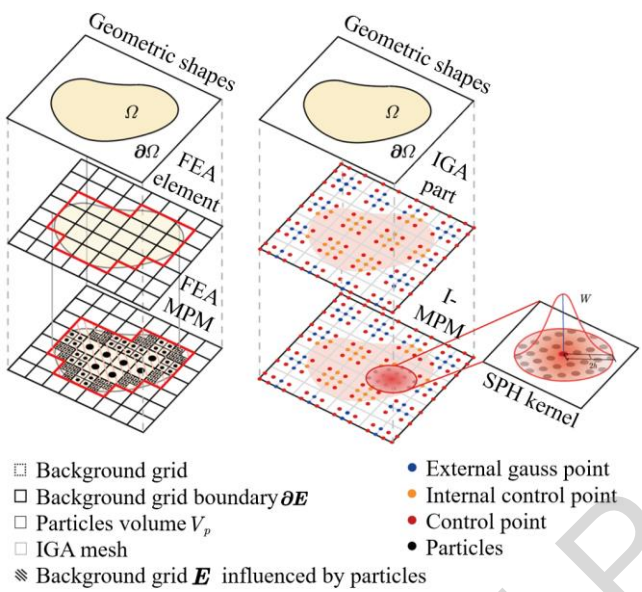

Fig. 2 Schematic diagram of the mapping comparison between the MPM and the I-MPM: (a) MPM, (b) I-MPM.

Under this Euler update mechanism, spatial discretization is implemented on a finite-dimensional subspace $\mathcal{V}^h \subset \mathcal{V}_v$ through the standard MPM method, where each particle $\boldsymbol{p}$ carries the corresponding weight or volume $\boldsymbol{V}_p$ and participates in the weighted mapping and integration of field variables. Because the midline basis functions in traditional finite element methods are not differentiable at mesh boundaries, they can cause "cell-crossing errors," affecting numerical accuracy. To improve the accuracy of static analysis, this paper uses a cubic B-spline kernel function based on SPH to replace the original kernel function, thereby effectively reducing the error and making the mapping process of particle physics information to the IGA grid smoother.

The specific SPH kernel function weights the particle physical quantities in the control point neighborhood, and the control point physical quantity information δ_c is calculated as follows:

$$\delta_c = \sum_{i=1}^{\alpha} \delta_p^i \boldsymbol{W}\left(\frac{\left|\boldsymbol{x}_p^i - \boldsymbol{x}_c\right|}{\boldsymbol{h}}\right) \boldsymbol{V}_\alpha, \qquad (9)$$

where $\boldsymbol{x}_p^i, \boldsymbol{x}_c$ are the particle and control point positions, δ_p^i is the material consumption of the *i*th particle, $\boldsymbol{w}$ is the cubic spline kernel function, and the kernel size $\boldsymbol{h}$ is related to the control point volume $\boldsymbol{V}_\alpha$. The specific form of the kernel function is:

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$$\boldsymbol{W}(\Theta)=\mathrm{Con}\cdot\begin{cases}1-\frac{3}{2}\Theta^2+\frac{3}{4}\Theta^3, & (0<\Theta<1)\\ \frac{1}{4}(2-\Theta)^3, & (1<\Theta<2)\\ 0, & \text{otherwise.}\end{cases} \tag{10}$$

The constant Con is $10/\left(7\pi\boldsymbol{h}^2\right)$ and $1/\left(\pi\boldsymbol{h}^3\right)$ in two-dimensional and three-dimensional space respectively. The above steps are called the "particle to control point (P2C)" stage, which provides control information for the subsequent Eulerian Lagrangian grid of I-MPM. Then, the control point information is re-transmitted to the particles using NURBS basis function mapping. NURBS basis functions have high-order continuity, which ensures smooth mapping of control point information and particle information after update. Based on the above geometric mapping and physical quantity approximation, the physical domain variables can be expressed as

$$\boldsymbol{u}_h(\boldsymbol{x})=\sum_{A=1}^{N_c}\boldsymbol{u}_A\boldsymbol{R}_A(\boldsymbol{x}), \tag{11}$$

where

$$\boldsymbol{R}_A(\boldsymbol{x})=\hat{\boldsymbol{R}}_A\left(\xi^{-1}(\boldsymbol{x})\right). \tag{12}$$

In the static analysis of the coupled IGA-MPM, particle motion and property evolution can be viewed as Lagrangian, while the smooth update of the control point density or stress field corresponds to an Eulerian description. A mapping relationship is established between the two descriptions via high-order NURBS basis functions to facilitate information transfer: in the particle-to-grid mapping (P2G) phase, the physical quantities carried by the particles are assigned to the control points in an Eulerian manner; in the grid-to-particle mapping (G2P) phase, the particle states are updated in a Lagrangian manner, thus maintaining the continuity and conservation of the physical field. This method combines the advantages of both Lagrangian and Eulerian descriptions, integrating step-by-step discretisation with spatial mapping, and utilising high-order kernel function projection to achieve high-precision simulation of material point motion and static fields. As shown in Fig. 2, Fig. 2(a) illustrates a typical representation of a continuum region in the traditional material point method (MPM). Unlike the traditional material point method (MPM), which relies solely on a fixed Euler mesh for linear approximation, the IGA-MPM uses NURBS basis functions to construct a high-order C^n continuous space. This maintains the smoothness and consistency of strain rate and stress fields as particles move across cells, significantly improving the stability and accuracy of structural response calculations. In traditional MPM, material points calculate continuous variables such as stress and displacement based on material coordinates, similar to finite element integration points. However, within the IGA framework, the number and distribution of material points within each cell can be adaptively adjusted based on analysis accuracy, achieving higher-resolution physical field representation [51,52].

Fig. 2(b) illustrates the I-MPM domain mapping process. This method tightly couples material points with control points through a smoothing kernel function and NURBS mapping, ensuring continuous and uniform distribution of density and stiffness fields. First, each particle carries a material density parameter, ρ_p, and a physical position, $\boldsymbol{x}_p$. This particle density information is mapped to a control point using the kernel function $\boldsymbol{W}\left(\boldsymbol{x}_p-\boldsymbol{x}_c\right)$, yielding the physical density $\rho_c=\sum_p\boldsymbol{W}(\cdots),\rho_p,\boldsymbol{V}_p$ at the control point. Subsequently, the density information of the control point is transferred to the background mesh using the NURBS basis functions $\boldsymbol{N}_i$, constructing a smooth and differentiable global stiffness matrix $\boldsymbol{K}(\Phi)$, ensuring continuity and numerical stability of the structural field. When solving the displacement field and performing sensitivity analysis, the physical quantities can be mapped back from the control points to the particles, achieving a continuous conversion between density, stiffness, and flexibility. Compared with traditional MPM, I-MPM not only avoids the numerical oscillation and stress noise problems caused by particles passing through cells, but also effectively suppresses the occurrence of local extreme softening and false hinges, allowing the topology optimized structure to have a controllable stiffness distribution while maintaining overall flexibility, thereby significantly improving the physical rationality, numerical robustness and manufacturing feasibility of the structural design.

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3.2 The Lagrangian-Eulerian particle flow model

This paper proposes a topology optimization framework for Lagrangian-Eulerian hybrid particle flow model based on IGA for flexible mechanisms [53]. This method integrates the IGA framework constructed by material point method (MPM), smoothed particle hydrodynamics (SPH) kernel function and NURBS basis function, effectively alleviating the numerical instability problem caused by unit intersection in traditional topology optimization, and improving the smoothness and accuracy of sensitivity analysis. This method is built on the synergy of three different description fields.

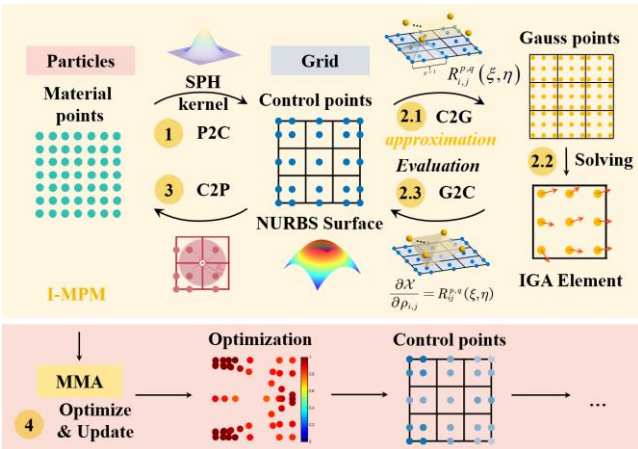

Fig. 3 The graphical flowchart of the Lagrangian-Eulerian particle flow model.

- Particle field: particles are movable design variables and are freely distributed in a unified background grid. Each particle carries the position and material consumption, collectively referred to as design variables;
- Control point field: the physical information of particles is mapped to control points through the SPH kernel function to obtain the aggregation variables of the control points. The variables are then numerically projected and converted into physical density on the control points to describe the material distribution;
- Gaussian point field: used for numerical integration and solving control equations. The physical density at the Gaussian point is obtained through the density distribution function (DDF), which is constructed by combining the control density and NURBS basis functions.

The flow chart of the Lagrangian-Eulerian particle flow model is shown in Fig. 3. The key computational procedure of the proposed method can be summarized as follows. First, particles are uniformly distributed on the IGA grid within the parameter space and mapped to the physical domain via NURBS. The initial material allocation is prescribed according to the target volume fraction to satisfy the global volume constraint, while all control volumes are simplified to unity for computational convenience [54]. Subsequently, the material information carried by particles is transferred to the control points through the SPH kernel function, forming aggregated variables at the control points. The corresponding physical density of the control points is then obtained via numerical projection, thereby completing the data conversion from a Lagrangian (particle) to an Eulerian (control point) description. On this basis, the IGA solution process is performed. Specifically, the physical density at the Gaussian points is interpolated from control point densities using NURBS basis functions, which ensures numerical stability and local adaptability. The density distribution function is given by:

$$\rho_g(\xi,\eta,\zeta)=\sum_{i=p-r}^{p}\sum_{j=q-s}^{q}\sum_{k=t-u}^{t}\boldsymbol{R}_{i,j,k}(\xi,\eta,\zeta)\rho_{i,j,k} \tag{13}$$

Based on this material distribution, a linear elastic system that satisfies the imposed boundary conditions is constructed and solved, yielding the displacement field $\boldsymbol{u}$. The optimization objective is to maximize the displacement response at a specified output location, where a displacement component $\boldsymbol{u}_{\text{out}}$ generated by the unit force load at the output node is often used as the objective function (for example, the displacement $\left|\boldsymbol{u}_{\text{out,x}}\right|$ along a certain direction). Sensitivity analysis is then performed: the results evaluated at Gaussian points are projected back to the control points using the NURBS basis functions, leading to the following expression for sensitivity with respect to the control variables:

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$$\frac{\partial \rho_p}{\partial \rho^{i,j,k}}(\xi,\eta,\zeta) = \boldsymbol{R}_{\varphi}^{i,j,k}(\xi,\eta,\zeta) \qquad (14)$$

The sensitivity information of the control points (the sensitivity of the objective function and constraint function to the design variables, i.e., $\mathrm{d}\boldsymbol{J}/\mathrm{d}\vartheta$ and $\mathrm{d}\boldsymbol{V}/\mathrm{d}\vartheta$) are subsequently mapped from the control points back to the particles through the SPH kernel function combined with numerical projection. This establishes the complete chain of derivatives with respect to the particle design variables, thereby providing the gradient information necessary for optimization. Finally, the Method of Moving Asymptotes (MMA) is employed to iteratively update particle material allocations, driving the structural topology evolution. The particle states are continuously updated, and the above procedure is repeated until convergence criteria are satisfied. The Lagrangian-Euler hybrid topology optimization method achieves precise control and optimization of material distribution through efficient information interaction between the design domain and the solution domain, combined with the geometric continuity of IGA and the numerical smoothness of the SPH kernel. When the optimization goal is to maximize the output displacement, this framework shows stronger numerical robustness and structural controllability in dealing with large deformation, mesh instability or explicit hinge problems, effectively avoiding the occurrence of actual hinge phenomenon.

4 The mathematical optimization formulation

4.1 The maximization of single point displacement

This paper proposes a compliant mechanism optimization method based on the Lagrangian-Euler particle flow model. The optimization goal is to maximize the performance index of the compliant mechanism by adjusting the position of particles, material density and structural parameters under the action of given input and output forces. The material density is weightedly mapped from the Lagrangian particle $\boldsymbol{x}_p$ to the control point $\boldsymbol{x}_c$ with the kernel function $\boldsymbol{W}$, which realizes the smooth transmission of physical information between particles and control points. The optimization formula is as follows:

$$\begin{cases} \text{Find: } \mathcal{V} = \left\{ \boldsymbol{x}^n, \delta_\rho^n \right\}, \\ \text{Min: } \boldsymbol{J} = \alpha \boldsymbol{F}_{\text{in}}^{\text{T}} \boldsymbol{U}_{\text{in}} + \beta \boldsymbol{F}_{\text{out}}^{\text{T}} \boldsymbol{U}_{\text{out}} - \boldsymbol{F}_{\text{out}}^{\text{T}}, \\ \text{s.t., } \begin{cases} \boldsymbol{K}(\Phi)\boldsymbol{U}_{\text{in}} = \boldsymbol{F}_{\text{in}}, \\ \boldsymbol{K}(\Phi)\boldsymbol{U}_{\text{out}} = \boldsymbol{F}_{\text{out}}, \\ \dfrac{1}{|\boldsymbol{\Omega}|}\displaystyle\int_{\boldsymbol{\Omega}} \Phi\left(\rho_e^{i,j,k}, \xi, \eta, \varsigma\right) \mathrm{d}\boldsymbol{\Omega} \le \boldsymbol{V}_{\text{lim}}, \\ 0 < \rho_e^{\min} \le \rho_e^{i,j,k} \le 1, \\ \Phi(\boldsymbol{\rho}) = \displaystyle\sum_p \boldsymbol{W}\left(\boldsymbol{x}_p - \boldsymbol{x}_c\right) \delta_\rho^p \boldsymbol{V}_p. \end{cases} \end{cases} \qquad (15)$$

The specific objective function is defined as $\boldsymbol{J} = \alpha \boldsymbol{F}_{\text{in}}^{\text{T}} \boldsymbol{U}_{\text{in}} + \beta \boldsymbol{F}_{\text{out}}^{\text{T}} \boldsymbol{U}_{\text{out}} - \boldsymbol{F}_{\text{out}}^{\text{T}}$. The static equilibrium condition of the structure is expressed by the stiffness matrix $\boldsymbol{K}(\Phi)$ based on the material density function $\boldsymbol{K}(\Phi)\boldsymbol{U}_{\text{in}} = \boldsymbol{F}_{\text{in}}$ and $\boldsymbol{K}(\Phi)\boldsymbol{U}_{\text{out}} = \boldsymbol{F}_{\text{out}}$ The material volume distribution is subject to the volume constraint and the material density satisfies $0 < \rho_e^{\min} \le \rho_e^{i,j,k} \le 1$ .Among them, the design parameters ε_i and $\boldsymbol{P}_{i,j}$ are both positive values. The material density function Φ is weightedly mapped from the Lagrangian particle $\boldsymbol{x}_p$ to the control point $\boldsymbol{x}_c$ through the kernel function $\boldsymbol{W}\left(\boldsymbol{x}_p - \boldsymbol{x}_c\right)$. The proposed particle-flow topology optimization method takes Lagrangian particles as the core design variables and directly controls the evolution of structural geometry and physical properties, thereby achieving coordinated optimization of material distribution, geometric shape and physical field. According to Ref. [20], the weighting factors α and β change during the optimization iteration and are updated by the following formula:

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$$\alpha^{k+1}=\begin{cases}0, & k=1,\\ \dfrac{F_{\text{out}}^{\mathrm{T}}U_{\text{in}}^{k}}{F_{\text{in}}^{\mathrm{T}}U_{\text{in}}^{k}}, & k\geq 2,\end{cases}\qquad \beta^{k+1}=\begin{cases}0, & k=1,\\ \dfrac{F_{\text{out}}^{\mathrm{T}}U_{\text{in}}^{k}}{F_{\text{out}}^{\mathrm{T}}U_{\text{out}}^{k}}, & k\geq 2,\end{cases}\tag{16}$$

where U_{in}^{k} and U_{out}^{k} represent the displacement vectors obtained at the kth iteration, and α^{k+1} and β^{k+1} represent the weight factors at the (k+1)th iteration. The continuous density field obtained by NURBS basis function interpolation is used to describe the distribution state of the material in the physical space. The expression of DDF is as follows:

$$\Phi(\xi,\eta,\zeta)=\sum_{i=1}^{N_\xi}\sum_{j=1}^{N_\eta}\sum_{k=1}^{N_\zeta}R_{i,j,k}^{r}(\xi,\eta,\zeta)\cdot\rho^{i,j,k},\tag{17}$$

where $R_{i,j,k}^{r}$ is a two-dimensional NURBS basis function, $\rho^{i,j,k}$ are density variables at control points. To ensure numerical stability, the density value is limited to a closed interval $\left[\rho^{\min},1\right]$, where $\rho^{\min}\ll 1$ is used as an approximate space instead of completely zero density. This construction method not only provides highly smooth geometric description capabilities, but also significantly reduces the probability of local rigid units and actual hinges in optimization.

The boundary is identified by density contours, and the constant contour value is $\Phi_0=0.5$, thus forming the following material classification categories:

$$\Phi(\xi,\eta,\zeta)=\begin{cases}>\Phi_0, & \text{solid},\\ =\Phi_0, & \text{boundary},\\ <\Phi_0, & \text{void}.\end{cases}\tag{18}$$

The density value of the control point comes from the Lagrangian particles distributed in the IGA parameter domain. Each particle carries a design variable δ_p, which represents the local material usage, and projects its information onto the control point through a weighted kernel function, thus forming a density field. The control point density is constructed as follows:

$$\rho^{i,j,k}=\sum_{p}R_{i,j,k}^{r}(\xi_p,\eta_p,\zeta_p)\cdot\rho_p(\delta_p).\tag{19}$$

To achieve the continuity and smoothness of the structural topology and avoid local singular rigidity or actual hinges caused by finite element discretization or mesh dependence, the particle density function $\rho_p(\delta_p)$ adopts a three-segment interpolation expression to achieve segment-by-segment continuous transition, as follows:

$$\rho_p(\delta_p)=\begin{cases}\tanh\left(1+c_0\delta_p\right)\cdot\delta_p & 0\leq\delta_p<p\\ 1-c_1\left(\delta_p-p\right)^3+1 & \delta_p\geq p\end{cases}\tag{20}$$

where c_0,c_1 are adjustment constants, and p is the limit of the control segmented smoothing region. This function has good numerical differentiability, which can maintain global continuity and avoid discontinuities or geometric sharp angles in optimization, thus theoretically suppressing the generation of "point flexible structures" or "fact hinges".

The structural response analysis is completed through the IGA unit stiffness matrix, which is expressed in physical space as follows:

$$K_e=\int_{\Omega_e}B^{\mathrm{T}}DB\cdot\rho_c(\xi,\eta,\zeta)\,\mathrm{d}\Omega_e,\tag{21}$$

where K_e is the unit stiffness matrix, B is the strain-displacement matrix, and D is the material stiffness matrix. The density function ρ_c is provided by the DDF, which ensures that the material distribution in the integration domain is continuous and smooth. The above integral is highly dispersed using the Gaussian integral in parameter space:

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$$\boldsymbol{K}_e = \sum_{g=1}^{n_g} \boldsymbol{B}^{\mathrm{T}} \cdot \Phi \cdot \boldsymbol{D}_0 \cdot \boldsymbol{B} \cdot |\boldsymbol{J}| \cdot \omega_\xi \omega_\eta \omega_\zeta . \qquad (22)$$

This density construction method has two major theoretical advantages: First, through NURBS The high-order continuity of the basis function controls the density distribution, so that a compliant structure with distributed flexibility can be constructed, avoiding the local softening behavior caused by stress concentration or sharp corner features; secondly, since the material variables are based on particles rather than fixed grid settings, the entire design domain no longer relies on the topological connection method of the Euler grid, fundamentally avoiding the de facto hinge problem caused by the "single node connection" form in the finite element method.

4.2 Design variable sensitivity analysis

In the optimization of compliant mechanisms based on maximizing single-point output displacement, the design variables usually include the particle position $\boldsymbol{x}_p^n$ and its corresponding material consumption δ_p^n. The sensitivity analysis of the objective function to these variables is derived using the chain rule, where the control point density ρ_c is used as an intermediate variable to connect the physical properties of the particle with the overall structural response, and is:

$$\frac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}\vartheta} = \frac{\partial \boldsymbol{J}}{\partial \rho_c} \cdot \frac{\partial \rho_c}{\partial \delta_\rho} \cdot \frac{\partial \delta_\rho}{\partial \vartheta}, \qquad (23)$$

where $\vartheta = \boldsymbol{x} * \boldsymbol{p}, \delta * \rho^{\boldsymbol{p}}$ represents the design parameters of the particle; ρ_c represents the material density at the control point, and the Φ function maps the material information of the particle to the Euler control point through the SPH kernel function:

$$\Phi(\rho) = \sum_p \boldsymbol{W}\left(\boldsymbol{x}_p - \boldsymbol{x}_c\right) \delta_\rho^p \boldsymbol{V}_p . \qquad (24)$$

The contribution of particles to the control point density is calculated by the SPH projection kernel $\boldsymbol{W}(\boldsymbol{r}/\boldsymbol{h})$. The piecewise differentiable representation is:

$$\frac{\partial \delta_c}{\partial \delta_p} = \boldsymbol{W}\left(\frac{\boldsymbol{r}}{\boldsymbol{h}}\right), \quad \frac{\partial \delta_c}{\partial \boldsymbol{x}_p} = \delta_p \cdot \frac{\mathrm{d}\boldsymbol{W}}{\mathrm{d}\boldsymbol{r}} \cdot \frac{\boldsymbol{x}_d}{\boldsymbol{r}} . \qquad (25)$$

The update of the control point density ρ_c is based on the projection function, and its derivative form with respect to the aggregate variable δ_ρ^c can be further obtained to control the smoothness of the density change process. The objective function is the displacement combined response of the compliant mechanism under the input and output loads:

$$\boldsymbol{J} = \alpha \boldsymbol{F}_{\mathrm{in}}^{\mathrm{T}} \boldsymbol{U}_{\mathrm{in}} + \beta \boldsymbol{F}_{\mathrm{out}}^{\mathrm{T}} \boldsymbol{U}_{\mathrm{out}} - \boldsymbol{F}_{\mathrm{out}}^{\mathrm{T}} \boldsymbol{U}_{\mathrm{in}} . \qquad (26)$$

In the context of IGA, the structural response (displacement) corresponding to the control point density can be solved by Gaussian integration. Finally, the sensitivity of the output displacement to the control point density $\rho^{i,j,k}$ can be obtained:

$$\frac{\partial \boldsymbol{J}}{\partial \rho_c^{i,j,k}} = \left[\boldsymbol{B}^{\mathrm{T}} \boldsymbol{D} \boldsymbol{\varepsilon}\left(\boldsymbol{u}_{\mathrm{out}}\right)\right]_{i,j,k}^{\mathrm{T}} \qquad (27)$$

and through the projection kernel and chain propagation back to the particle design variables, the complete composition is:

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$$\begin{cases} \dfrac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}\delta_p} = \sum\limits_{i,j,k} \dfrac{\partial \boldsymbol{J}}{\partial \rho_c^{i,j,k}} \cdot \dfrac{\partial \rho_c^{i,j,k}}{\partial \delta_p}, \\ \dfrac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}\boldsymbol{x}_p} = \sum\limits_{i,j,k} \dfrac{\partial \boldsymbol{J}}{\partial \rho_c^{i,j,k}} \cdot \dfrac{\partial \rho_c^{i,j,k}}{\partial \boldsymbol{x}_p}. \end{cases} \tag{28}$$

This expression starts from displacement sensitivity and uses SPH kernel function and NURBS interpolation throughout the process to achieve smooth coupling between particles, control points and structural responses, and ensure that sensitive information is not broken. In theory, it can avoid the formation of local rigidity or "factual hinges" and improve the smoothness and convergence performance of the entire optimization process.

5 Numerical implementations

The proposed particle-flow topology optimization (PFTO) method integrates Lagrangian particles with an Eulerian IGA grid and uses MMA to iteratively update design variables. Particles store position and material information and move freely, while structural responses are computed on the fixed grid. Particle data are mapped to control points through an SPH kernel to form a density field, which—combined with NURBS basis functions—constructs the structural topology; the density is further transferred to Gauss points for stiffness evaluation. Sensitivities are propagated from Gauss points to control points and particles to complete the design update. This framework effectively merges particle flexibility with grid-based accuracy, ensuring both robustness and adaptability in topology optimization. In addition, the method maintains numerical consistency across mappings, enhancing the overall stability of the optimization process. The implementation steps of the PFTO method are illustrated in Fig. 4.

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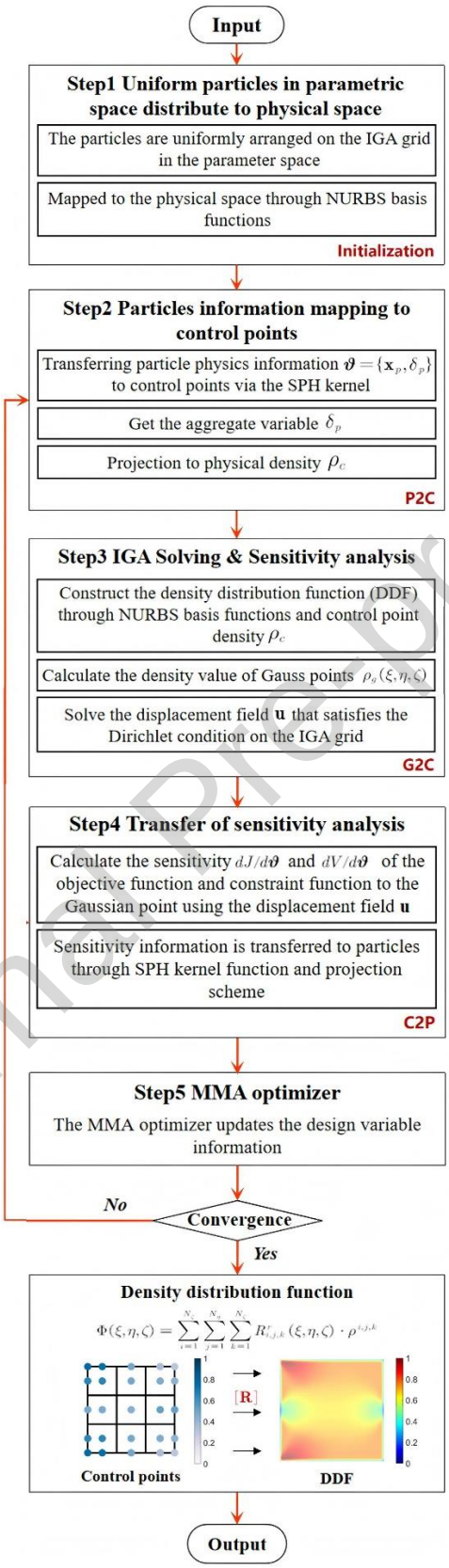

Fig. 4 Flowchart of the PFTO approach.

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In this section, the main purpose is to study the effectiveness, superiority and indispensability of the proposed particle-flow topology optimization (PFTO) method by fully discussing several numerical examples of 2D and 3D compliant mechanisms. First, the PFTO method is used to optimize a classic two-dimensional cantilever beam to demonstrate its basic characteristics and verify its effectiveness in improving structural performance through iterative evolution of structural topology, especially showing the unique characteristics of particle motion in the design domain. The relevant structural parameters in the numerical examples remain consistent, that is, Young's modulus $\boldsymbol{E}_0 = 1, \boldsymbol{E}_{\min} = 1.0\times10^{-9}$, Poisson's ratio $\boldsymbol{v} = 0.3$, and the penalty parameter $\boldsymbol{p_e}$ in the formula is set to 3. In all numerical cases, the magnitude of the applied load is defined as 1. The convergence condition of the optimization process is that the optimization is considered to have converged when the $\boldsymbol{L}_\infty$ norm of the objective function changes less than 0.1% in consecutive iterations or reaches a maximum of 150 steps.

5.1 Displacement inverter

5.1.1 Basic characteristics of the PFTO method

This section aims to demonstrate the basic characteristics of the proposed PFTO method in the design problem of maximizing the classical single-point displacement output of flexible mechanisms. Taking the classical displacement inverter as the research object, the focus is on discussing and comparing the PFTO method's effect on suppressing the "hinge" problem during the optimization process, thereby revealing its advantages over traditional topology optimization methods.

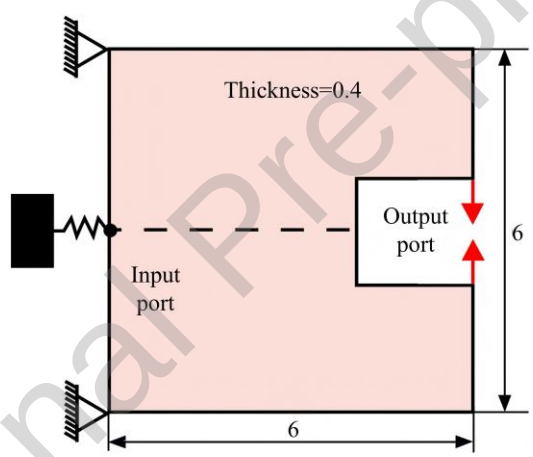

Fig. 5 Compliance mechanism synthesis problem for displacement inverters.

It is worth noting that the displacement inverter is a typical mechanism used to change the driving force or displacement direction, and its optimization design method has become one of the most representative benchmark problems in the field of flexible mechanism topology optimization Fig. 5 shows the size of the design domain and the external load. The boundary size $L = 5, H = 4$, the upper and lower ends of the left boundary are fixed, and the input displacement from the left midpoint to the right is converted into the output displacement from the right midpoint to the left. The number of isogeometric units is divided 100×80, and the node-wise topology optimization design model in the two parameter directions is defined respectively. This problem uses the weighting function $\boldsymbol{J}$ in Eq. (23) as the objective function, and the weight factors α and β are adaptively adjusted according to Eq. (24).

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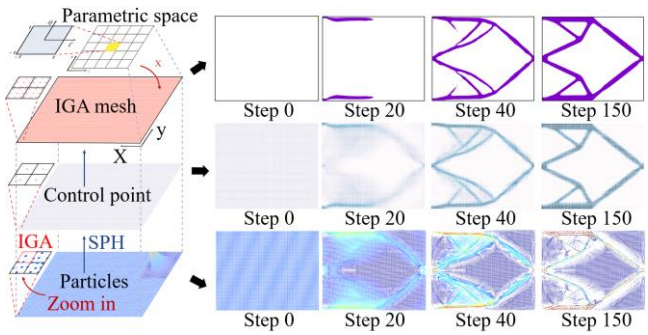

Fig. 6 Initial distribution and iteration of particles, control points, and Gaussian quadrature points in the design domain.

As shown on the left side of Fig. 6, the initial distribution of control points, particles, and Gaussian integration points in the design domain is first shown. To more clearly illustrate the relationship between the three, a layered exploded diagram is further used to illustrate the specific arrangement of different point types within each IGA cell. The control points remain fixed in the Euler coordinate system, and their physical density is obtained from the particles through a mapping process. This mapping is based on the smoothed particle hydrodynamics (SPH) kernel function (Eq. (18)) and operates within a neighborhood with a radius of twice the cell size, converting the particle state information into aggregated variables of the control points (Eq. (17)). These density variables are then transferred to the Gaussian integration points for use in the displacement field solution in the IGA numerical calculation. The right half of Fig. 6 shows the typical evolution of the displacement inverter at different iteration stages (iterations 0, 20, 40, and the final converged result). The second row shows the distribution of the physical density of the control points, where the dark cyan region corresponds to a solid material layout with a density of 1. The third row provides the spatial distribution of the particles and their material consumption. As iterations progress, particles gradually converge toward critical force paths, increasing the density of the generated point neighborhood. Driven by displacement sensitivity, the particle distribution further concentrates toward structural components. Overall, the simultaneous evolution of particle motion and material consumption clearly depicts the dynamic process of structural topology from initial formation to final shape.

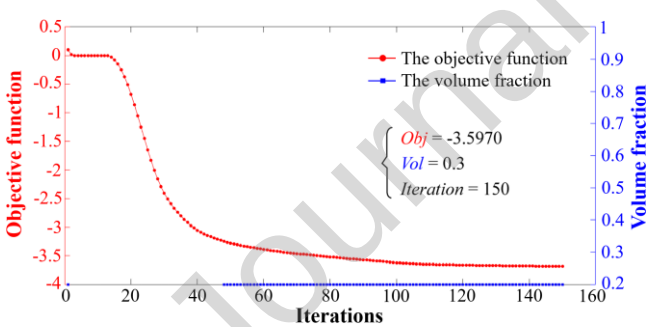

Fig. 7 Convergent histories.

During the iterative topology optimization process, particles, acting as design variables, undergo position updates within a Lagrangian framework, driving the evolution of material distribution. Unlike fixed control points, particles can freely migrate within the design domain, and their movement directly determines the accumulation of material along critical load-bearing paths. To ensure rational material utilization, a numerical projection method normalizes the physical density derived from particle mapping. This not only improves the efficiency of material distribution but also enhances the optimization effect of single-point displacement output. Fig. 7 shows the iterative process of the displacement inverter objective function and volume fraction, showing that both convergence curves exhibit extremely high smoothness and stability. During the first 40 iterations of the optimization, the structural topology rapidly and significantly adjusted, forming an intermediate configuration with good performance. Subsequent iterations focused on refining and fine-tuning details to further improve overall performance. The entire optimization process terminated at step 150, demonstrating the outstanding efficiency of the proposed PFTO method. The optimization results include not only the positional evolution of the particles within the design

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domain and the corresponding material consumption distribution, but also the physical density of the control points and the resulting structural topology with clear boundaries and distinct solid-void demarcations. These features together verify the effectiveness of this method in improving the bearing capacity of the structure.

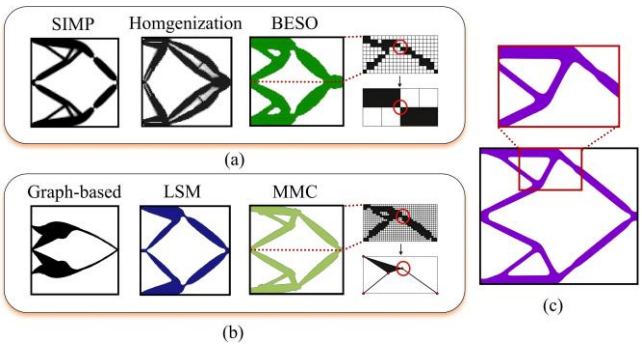

Fig. 8 Comparison between PFTO and classic optimization methods for displacement Inverters: (a) Mesh-based topology optimization methods, (b) Topology optimization method based on geometric representation, (c) PFTO

It's important to emphasize that the smoothness and continuity of the topological results stem from two mechanisms: first, the material representation is based on NURBS functions, whose geometric descriptions are inherently smooth; second, the particle-to-control point mapping implemented within the I-MPM framework using the SPH kernel ensures the continuity of the control point density field. Particle density contributes to the control points in a weighted manner within its support domain, while the kernel radius determines the local range of influence, effectively maintaining the smoothness of the density field globally. This modeling and optimization strategy avoids the numerical precision and convergence instability common in traditional finite element methods, achieving a high degree of consistency between the material representation model and the numerical analysis model. This consistency not only eliminates the adverse effects of low precision but also ensures stable convergence of the optimization process, ultimately resulting in a design with superior performance. The comparative results in Fig. 8 further demonstrate that traditional mesh-based topology optimization methods (such as SIMP and ESO) often generate extremely soft regions with single-node connections at element corners when dealing with flexible mechanisms. This is known as the "de facto hinge" problem. This hinge phenomenon leads to localized insufficient stiffness, disrupts structural continuity, and adversely affects practical manufacturability. Although previous studies have attempted to mitigate this problem through filters, projection functions, or post-processing smoothing, these methods generally only improve local geometric features and fail to fundamentally eliminate hinge formation.

5.1.2 Discussions on the SPH core

This section focuses on the impact of the SPH kernel radius on the topology optimization results of flexible mechanisms within the I-MPM framework, particularly its role in suppressing hinging. The SPH kernel radius is based on the element size, and its influence domain is defined by the kernel coefficient. Within this radius, particles perform a weighted projection of physical quantities on control points, thereby determining the continuity and smoothness of the control point density field. As shown in Fig. 9, the kernel function curves corresponding to different radii exhibit a clear coupling relationship with the IGA physical mesh scale, a feature that directly affects the smoothness of the particle-to-control point mapping. It can be observed that as the SPH kernel radius decreases, the range of influence of the particles on the control points decreases, resulting in a decrease in the density of the control points after the particles of the same density are mapped. To maintain a consistent distribution of control point density, the density of the particles themselves must be increased accordingly. Therefore, at smaller kernel radii, the particle distribution in the optimization results appears more sparse.

To systematically evaluate its effect, this study selected five different kernel radii (0.5, 1.25, 1.5, 1.75, and 3.5) and conducted numerical experiments. The numerical results, shown in Fig. 9, show that as the iterations progress, particles gradually migrate and form the primary force transmission paths: particles with high material consumption (red) concentrate towards the main load-bearing areas of the structure, while particles with low material consumption (blue) only move slightly near their initial positions. Under the overall constraints of the DDF method, the topologies obtained in most cases exhibit smooth boundaries and clear solid-void interfaces. However, when the core radius is too small (e.g., 0.5), the particle

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mapping range is extremely limited, resulting in highly discontinuous local density fields, noticeable jagged and sharp features at the topological boundaries, and the appearance of slender, fragile rods. This ultimately induces extremely soft connections similar to "de facto hinges," severely weakening load-bearing performance. Further comparison shows that when the radius is 1.25, 1.5, and 1.75, the optimization results are consistent with the baseline displacement inverter design, with objective functions of –3.5189, –3.5970, and –3.6675, respectively. This indicates that within this range, the core radius achieves an optimal balance between boundary smoothness and structural performance, effectively avoiding the appearance of hinges. In contrast, when the radius is increased to 3.5, although the boundary remains continuous, the excessively large influence domain leads to significant coarsening of the rods in the topology, unfavorable deformation at the joints, and the loss of the basic configuration. The objective function rises to –3.8791, the overall stiffness decreases, and the computational complexity increases significantly.

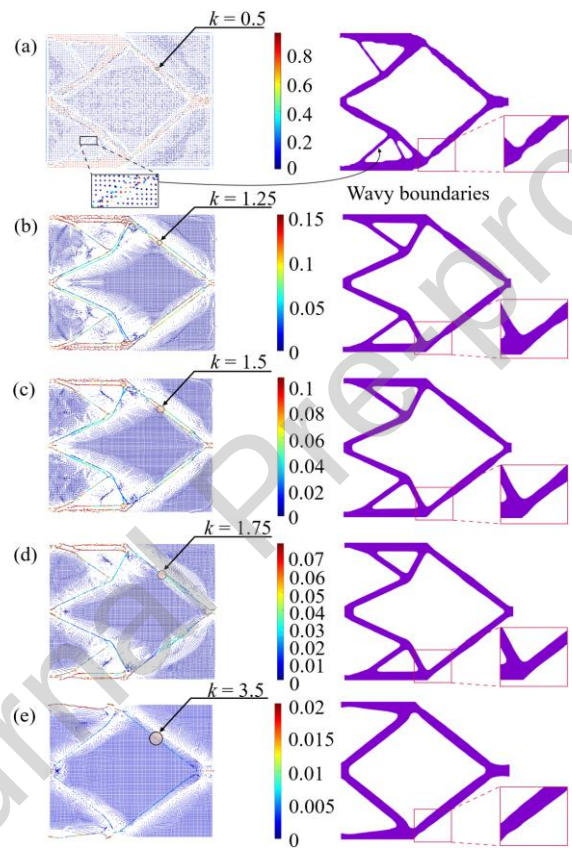

Fig. 9 Optimization results of displacement inverter with different radii in SPH kernel.

In summary, the SPH kernel radius plays a crucial role in PFTO results within the I-MPM framework. A too-small radius can disrupt mapping smoothness, resulting in localized fragile features and hinge effects. A too large radius introduces intermediate density, degrading performance and increasing computational overhead. Comprehensive analysis indicates that a reasonable range of values should be between (1, 2). In subsequent studies, unless otherwise specified, a default value of 1.5 will be used to balance performance, convergence, and numerical efficiency.

5.1.3 Discussions on particle mobility

This section aims to clarify the mobility of particles and point out the key role of initial particle layout in particle motion during the subsequent optimization process, which in turn significantly affects the flow characteristics of particles forming load transfer paths within the design domain. Two different particle distributions were defined within the design domain. All particles were randomly distributed within the design domain with varying densities. Using the nodes in the IGA mesh as references, two different particle distribution spacings were considered: 1 unit span and 1/2 unit span in parameter space.

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Fig. 10 illustrates the two different particle distributions within the design domain. It is readily apparent that the particles in the second distribution have a higher density than those in the first. It is noted that while the physical positions of the particles are randomly defined within the design domain, the particle material consumption values are consistent for both initial distributions. Under the specified maximum volume fraction definition, the detailed particle material consumption values for the corresponding initial conditions can be calculated by inverting the P2C mapping. It is also important to note that, except for the particle distribution density, all other design parameters remain the same in both cases.

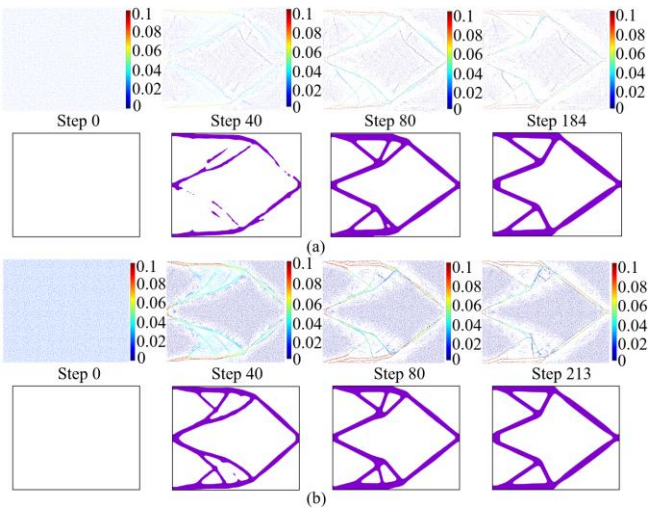

Fig. 10 Optimization results of displacement inverter under random particle distribution: (a) Initial random distribution of particles with a span of 1 element, (b) Initial random distribution of particles with a span of 1/2 element.

Fig. 10 also shows the final optimization results for the particles with different initial distribution spacings, as well as several intermediate results for optimizing the topology and particle distribution (including physical density and material consumption) for the first and second design cases. The figures clearly show that both design cases produce reasonable topologies, which closely resemble the baseline design results for a classic displacement inverter, regardless of whether the particle density within the design domain is increased or decreased. However, in the intermediate steps of Fig. 10(a), it is apparent that the geometry exhibits asymmetric design features. However, as the number of optimization steps increases and the particles continue to move, the topology remains consistent with the baseline. As the particle density within the design domain increases, as shown in Fig. 10(b), the intermediate steps become more stable and symmetrical when the particle spacing in parameter space is 1/2 the unit span. This indicates that the design domain contains sufficient particles carrying physical information, which can be robustly moved to the desired locations, compensating for the asymmetry of the initial particle layout. This key feature of particle movement enables the symmetrical generation of load transfer paths within the design domain, and the iterative process of the particles can be clearly observed in Fig. 10. Therefore, we can conclude that the PFTO (particle-based topology optimization) method is effective and robust in finding optimal topologies, even with a random initial particle distribution. Therefore, the mobility of particles effectively suppresses the numerical oscillations caused by discrete mapping errors during the optimization process, maintains the continuity and smoothness of physical quantities (such as density field, stress field and displacement field), and thus further improves the stability and accuracy of the optimization solution.

5.2 Push-type gripper

5.2.1 One basic characteristics of the PFTO method

In Fig. 11 shows the optimization object and boundary conditions of the gripper mechanism. The boundary size is $L=6, H=6$. The upper and lower ends on the left are fixed, and a rightward force is applied to the lower left end to generate a downward displacement at the upper right corner. The number of equal geometry units is 120×120. The node vectors Φ and Λ in the two parameter directions are defined as $\{0,0,0.000834,0.99167,1,1...\}$ and $\{0,0,0.01,...,0.0125,1,1\}$. The degree of the NURBS basis function is $\boldsymbol{p}=\boldsymbol{q}=1$. The number of control vertices is $\boldsymbol{n}=121, \boldsymbol{m}=121$.

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Similarly, the structural optimization design of the flexible gripper is performed by using the isogeometric method and the particle-flow topology optimization proposed in this paper. In the IGA method, the design domain is represented by a fixed NURBS grid; in the PFTO method, the design domain is composed of particles with initial distribution, and the filter radius $r_{\min}$=2 is set, and the particles can move freely during the optimization process. For the topology optimization of flexible grippers, the PFTO method demonstrates significant advantages compared to other classic topology optimization methods. While traditional optimization methods, such as the level set method and the moving shape method, can improve boundary smoothness, they still suffer from hinge effects and localized, extremely soft connections when dealing with structures like flexible grippers, especially when the structure evolves to complex shapes. These methods are prone to producing incoherent connections, which compromise the overall stiffness and gripping stability of the structure.

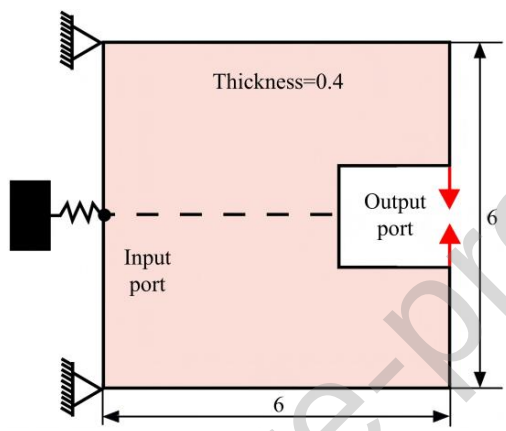

Fig. 11 Compliance mechanism synthesis problem for gripper mechanism.

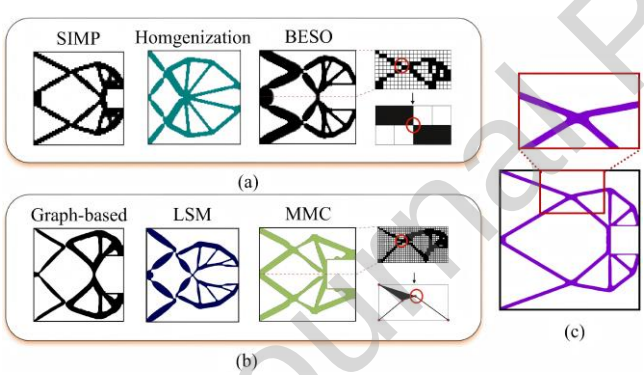

Fig. 12 Comparison between PFTO and classic optimization methods for flexible gripper: (a) Mesh-based topology optimization methods, (b) Topology optimization method based on geometric representation, (c) PFTO.

The PFTO method, by utilizing particle control point mapping and a weighted SPH kernel function, effectively avoids these drawbacks by ensuring global smoothness and continuity throughout the optimization process. Specifically, PFTO achieves a clearer interface between solids and voids in the optimization of flexible grippers, resulting in smoother topological boundaries and no "hinge" effects common with traditional methods. The results in Fig. 12 fully demonstrate the advantages of PFTO in this area, improving the structural stiffness and stability while ensuring the manufacturability and high performance of the optimized results. These advantages make the PFTO method ideal for flexible gripper design. On this basis, this study further investigated the influence of different SPH core radii on the topology optimization results of the flexible gripper. Previously, we can clearly see that the optimized topology shown in Fig. 9 is consistent with the design results provided in the previous article [45], which proves the effectiveness of the proposed PFTO method in other design fields.

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5.2.2 Discussions on the SPH core

Similar to the previous example, five core radii (0.5, 1.25, 1.5, 1.75 and 3.5) were selected for comparative analysis, and the numerical results are shown in Fig. 13.

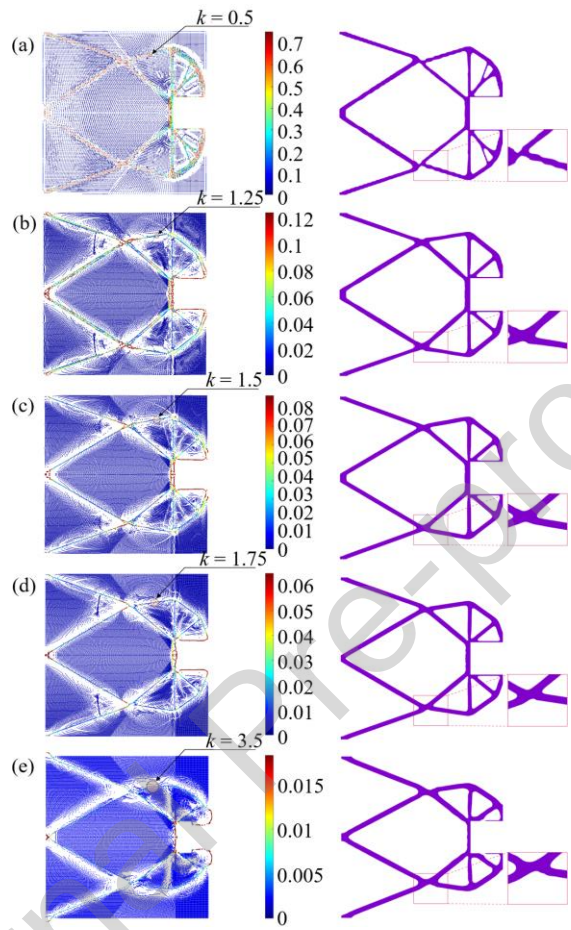

Fig. 13 Comparison between PFTO and classic optimization methods for flexible gripper.

The overall trend is highly consistent with the displacement inverter: as the iteration progresses, the particles gradually migrate and gather near the target clamping path, forming a clear force channel and functional configuration. When the core radius is too small (such as 0.5), the particle mapping domain is limited, resulting in severe discontinuity of the local density field, jagged and sharp features appearing on the clamping arm boundary, forming a slender and weak rod-like structure, and the clamping stiffness is significantly reduced, and even a "de facto hinge" appears. In contrast, when the core radius is in the range of 1.25–1.75, the resulting topology exhibits smooth boundaries and a reasonable clamping configuration, with objective function values (−0.5353, −0.5328, and −0.5244, respectively) close to the baseline design. This indicates that the core radius within this range effectively balances boundary smoothness with structural performance, thereby suppressing the hinge effect. However, when the core radius is further increased to 3.5, the topological boundary continuity deteriorates and breaks. The excessively large influence domain leads to excessive coarsening of the gripper arm members, resulting in unfavorable deformation at local nodes, resulting in reduced functionality and a deterioration of the objective function to −0.3137, while also increasing computational overhead.

5.2.3 Discussion on particle mobility

In the design of the flexible gripper, we employed two different particle distributions: a standard distribution with a spacing of 1 unit span and a high-density distribution with a spacing of 1/2 unit span. Fig. 14 shows the optimization results for both distributions. Regardless of the particle density, the final flexible gripper structure is highly consistent with the classical benchmark design. These results are consistent with the optimization results of the displacement inverter, further

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demonstrating that the PFTO method effectively optimizes the topology regardless of the particle distribution within the design domain, eliminating the randomness and asymmetry of the initial layout.

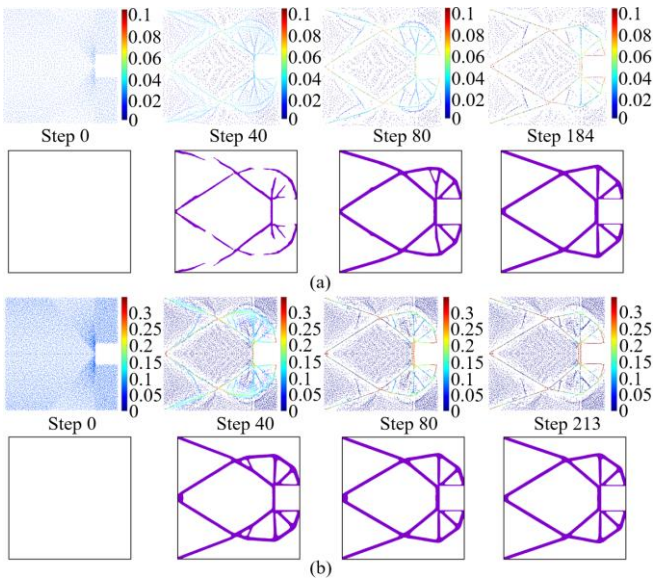

Fig. 14 Optimization results of displacement inverter under random particle distribution.

Through iterative particle motion, the load transfer path within the design domain is gradually and precisely constructed, ensuring the stability and functionality of the structural performance. Especially at higher particle densities (Fig. 14(b)), the optimization process exhibits greater numerical stability, with the geometry rapidly approaching symmetry in the early iterations and ultimately converging to a stable topology. This demonstrates that higher particle densities provide more precise load transfer paths and accelerate the restoration of symmetry during the optimization process, consistent with the optimization principles of the displacement inverter, thus verifying the reliability and adaptability of the PFTO method in various design scenarios.

It is worth noting that the mobility of particles plays a key role in the optimization process. During the iteration process, particles can freely follow the load flow path for adaptive migration, which enables the continuous transfer of physical quantities between the Euler and Lagrangian descriptions, thereby effectively suppressing the common numerical oscillation problems in traditional methods and maintaining the continuity and smoothness of the density field, stress field and displacement field. This dynamic redistribution mechanism based on particle movement enables PFTO to maintain stable solution performance and good topological consistency under complex geometric and non-uniform load conditions. In addition, the particle motion is driven by sensitivity analysis of the objective function and constraint function, which further ensures that the optimizer evolves in a physically reasonable direction, thereby improving the accuracy and physical interpretability of the structural design.

5.3 3D numerical designs

This section aims to demonstrate the applicability and advantages of the proposed PFTO method for three-dimensional design domain optimization. Two benchmark examples, a 3D displacement inverter and a 3D flexible gripper, are examined. In terms of numerical settings, particles are uniformly distributed throughout the design domain with the same material consumption, and the maximum material consumption ratio is limited to 30%. The initial particle spacing is equal to one parameter unit span, the SPH kernel radius in I-MPM is 1.5, and the initial material consumption of the particles is calculated by inverse mapping the particles to the control points. For the 3D displacement inverter structure, the design domain is a rectangular region with dimensions of 5×4×0.5. Fig. 15 illustrates the optimization process and initial conditions for the 3D Michell structure, including the particle distribution at four intermediate stages, highlighting the physical positions of the particles, material consumption, and the optimized topology.

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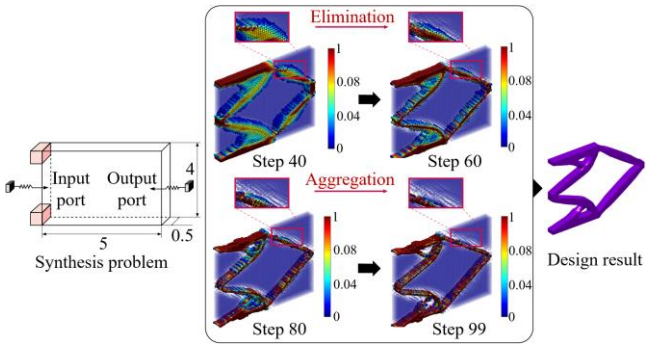

Fig. 15 Topology optimization design of 3D displacement inverter based on PFTO.

To better illustrate the particle motion characteristics within the design domain, four key particles are highlighted in red. First, the optimized 3D Michell structure is highly consistent with the results of previous studies under the same loads and boundary conditions, validating the effectiveness of the PFTO method for 3D Michell structure optimization. Furthermore, the optimized topology exhibits smooth and continuous boundaries. This is due to the use of NURBS to construct the DDF, which ensures high-order continuity and can accurately model any design domain. The material interface between the solid and void is also accurately preserved. Furthermore, during the optimization process, the particles are iteratively moved within the design domain to continuously optimize their distribution to maximize structural performance. The four key particles highlighted in red clearly demonstrate the evolution of particle motion, and the increase in particle material consumption is also reflected by the color change. This process demonstrates the effectiveness of the PFTO method in a three-dimensional design domain, adjusting material distribution through particle movement to enhance the structure's load-bearing capacity. The particles not only change their physical positions but also optimize and evolve reasonable material consumption, ultimately forming the structure's primary support. This demonstrates the necessity and advantages of Lagrangian descriptions in structural optimization.

Fig. 16 provides the convergence history of structural compliance and material volume fraction. Both compliance and material volume fraction converge smoothly, achieving a structure with the desired performance in only approximately 95 iterations, validating the efficiency of PFTO in 3D optimization. For a 3D flexible gripper, the optimization results, showing the constraints, the overall particle distribution, including particle physical positions, material consumption, and the optimized topology, are shown in Fig. 17. Identical particles can freely migrate throughout the 3D design domain according to the stress distribution and target displacement path, gradually forming a 3D component structure with clear functional partitioning and a continuous loading path. Similar to the two-dimensional case, particles tend to aggregate from non-critical regions toward critical stress regions during the optimization process, thereby achieving adaptive material distribution and ultimately forming a reasonable, continuous three-dimensional topology.

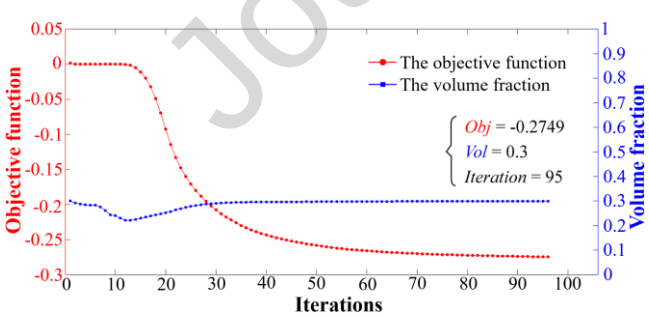

Fig. 16 Convergent histories.

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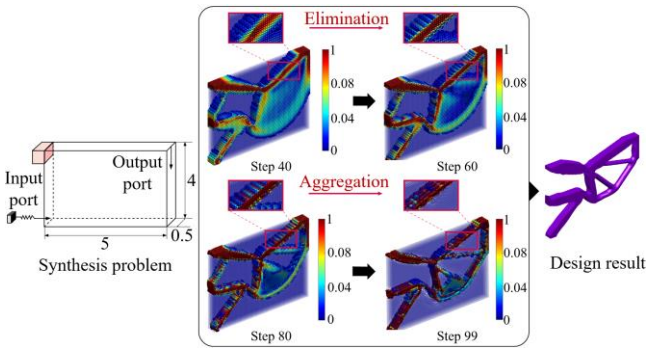

Fig. 17 Topology optimization design of 3D flexible gripper based on PFTO.

It is worth noting that traditional fixed-grid IGA topology optimization methods often face issues such as isolated material points, weak connections, and spatial hinges in the design of three-dimensional flexible mechanisms, and lack flexibility and stability when handling complex multiaxial deformations. In contrast, the PFTO method, based on a Lagrangian-Eulerian hybrid framework, utilizes particle mobility to retain the advantages of IGA geometric continuity while significantly enhancing its adaptability to three-dimensional multi-directional deformations and stress concentrations. In summary, the three-dimensional numerical results demonstrate the universality and stability of the PFTO method for the topology optimization of flexible mechanisms, demonstrating its applicability not only to two-dimensional structural design but also its potential for extension to the optimization of higher-dimensional complex structures.

6 Conclusions

This paper proposes a particle-flow topology optimization framework (PFTO) based on Lagrangian-Eulerian description suitable for the design of flexible mechanisms. By introducing the dual description of particle motion mechanism and field variables, the structural response ability and engineering applicability of flexible mechanisms under complex working conditions are effectively improved. This method combines the numerical stability of the material point method (MPM) with the geometric adjustability of particle flow, and realizes the continuous evolution of material distribution and adaptive adjustment of configuration under a unified topology optimization framework.

Specifically, on the one hand, the information coupling mechanism of Lagrangian particles and Euler background grid adopted by MPM significantly improves the physical stability and solution accuracy of flexible structures during large deformation; On the other hand, the free movement of particles breaks through the limitations of traditional fixed control points, so that the structural topology can dynamically respond to the load path and strain energy distribution during the evolution process, thereby generating a more reasonable and manufacturable compliant configuration. Through the particle aggregation and sparse behaviour during the optimization process, the material can be adaptively strengthened in the key area and eliminated in the non-key area, effectively meeting the dual requirements of deformation accuracy and force transmission continuity of the flexible mechanism. The 2D and 3D examples of several typical compliant mechanisms verify the versatility and robustness of the PFTO method. The results show that this method can not only effectively avoid the hinge degradation problem commonly seen in traditional methods, but also generate high-quality structural configurations with good mechanical properties and manufacturing feasibility. Based on its structural expression flexibility and the stability of the optimization process, this method is expected to be further expanded to the design of higher-dimensional and more complex flexible systems such as multi-field coupling, intelligent material-driven structures, and additive manufacturing path planning.

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CRediT authorship contribution statement

Jiayi Zhu was responsible for the overall methodology design and data analysis, conducted the investigation, validated the

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results, and wrote the original manuscript; Daji Lin assisted with data preparation, formal analysis, and provided necessary resources; Liang Gao supervised the project and was responsible for project administration, methodological guidance, and funding acquisition; Huai Cao contributed to conceptualization, conducted part of the investigation and formal analysis, and provided essential resources; Jie Gao led the conceptual framework, contributed to methodology and investigation, acquired funding, supervised the research, and revised the manuscript. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare no competing financial interests.

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Jiayi Zhu, born in 2000, is currently a doctoral candidate at *School of Aeronautics and Astronautics, Huazhong University of Science and Technology, China*. She received her bachelor's degree from *Huazhong University of Science and Technology, China*, in 2020. Her research interests include topology optimization.
Daji Lin, born in 2001, is currently a PhD candidate at *School of Mechanical Science and Engineering, Huazhong University of Science and Technology, China*. He received his bachelor degree from *Xidian University, China*, in 2019.
Liang Gao, born in 1974, is currently the Vice President of *Huazhong University of Science and Technology* and a professor at *School of Mechanical Science and Engineering*. He received his doctoral degree from *Huazhong University of Science and Technology, China*, in 2002.
Huai Cao, born in 1974, He is currently the Vice Dean and a professor of *School of Design, Huazhong University of Science and Technology, China*.
Jie Gao, born in 1993, is currently an associate professor at *School of Aeronautics and Astronautics, Huazhong University of Science and Technology, China*. He received his Ph.D. from *Huazhong University of Science and Technology, China*, in 2019.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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